AMD DEVELOPER HACKATHON · ACT III · TRACK 1 INTELLIGENT INDUSTRY

ghost expert

Your expert films a job once. Every new hire gets coached by her, live, and stopped before they skip a safety step.

GEMMA 3 · SIGLIP · WHISPER · KOKORO — ONE AMD INSTINCT GPU

THE PROBLEM

your best technician retires next year.

Thirty years of judgment

Which step people skip, and which skip gets them hurt. None of it is in the manual.

Training means standing there

Safety training is mandatory (in Italy, D.Lgs. 81/2008). Today a senior technician stands next to every new hire.

WHO USES IT

three people, three screens

AUTHOR VIEW

the expert

Films the job once, narrating. Fixes the AI's draft and approves it. Owns the procedure.

COACH VIEW

the trainee

Does the real job under a camera. Hears the next step, gets stopped before a dangerous skip, asks questions out loud.

TRAINING RECORD

the manager

Gets time per step against the expert, the mistakes caught, and a Competent or Needs another run verdict.

HOW IT WORKS

film once, coach everyone

1

Author

Whisper, SigLIP and Gemma cut the narrated video into 8 steps, each with a keyframe, a visual check and a safety flag.

2

Coach

4 frames a second from a top-down camera. Steps turn green as Gemma confirms them; a voice says what's next.

3

Stop and record

A skipped safety step stops the trainee. Everything lands in a printable training record.

THE MOMENT THAT MATTERS

stop. you haven’t done step 4.

The trainee reaches for the cover screws without testing that the machine is dead. The screen goes red, the voice stops them, and the expert’s own hands doing step 4 fill half the screen until they do it.

WHY IT ISN’T JUST A PROMPT

the model never decides alone

EVERY FRAME · ~10 MS

SigLIP

Guesses which step this looks like, only to decide when to ask Gemma. Cheap, and allowed to be wrong.

4 FRAMES + KEYFRAME

Gemma 3

Checks the guessed step and the newest skipped step in parallel: one round trip on the stop path.

OWNS THE STATE

Rule engine

A stop needs two agreeing checks on different frames. Unsure, it asks. Model down, it stays quiet.

STRESS TEST · SIMULATION

false stops: 14% → 0%

With Gemma wrong 5% of the time	One answer decides	Ghost Expert
Correct trainees falsely stopped	14%	0%
Skipped step 4 caught	98%	92%
Stop latency p50	0.8 s	1.6 s
Stop latency p95	1.8 s	2.8 s

We trade a little speed and catch rate for never shouting at a trainee who did it right.

SOURCE: EVAL/SIM_RESULTS.MD · 200 CLEAN + 200 SKIP-4 RUNS · SIGLIP WRONG ON 25% OF FRAMES · A DESIGN CHECK, NOT A GEMMA MEASUREMENT

POWERED BY AMD

four models, one gpu, no swapping

AMD Instinct MI300X — ROCM · VLLM · AMD DEVELOPER CLOUD

Gemma 3

12B vision, on vLLM. Verifies steps, drafts procedures, answers questions.

SigLIP

Embeds every frame against the expert's steps. Finds scene cuts.

Whisper

Narration with word timing. Trainee questions by voice.

Kokoro

82M voice. Speaks the coaching in English and Italian.

BEYOND THE STOP

her hands answer your question

Ask the expert

Hold to ask “where does the filter go?” The answer is spoken, and the expert’s clip of that moment plays.

Bilingual

Expert recorded in English, trainee coached in Italian, or the other way round.

Training record

Competent only with zero safety stops and zero overrides. Print to PDF or download JSON.

HARD QUESTIONS

what judges will ask

Question	Answer
“It’s just a vision model with a prompt.”	A deterministic rule engine owns safety. Two checks must agree, and we publish the eval.
“What if it misses a mistake?”	It’s a trainer, not a safety system. The lock is still the safety system. We show the miss rate.
“A camera filming workers, in Italy?”	Training only, started by the trainee. Frames stay in memory, not stored by default (Statuto dei Lavoratori, art. 4).
“Works on a toy box, not a plant.”	Authoring is general. A pilot means filming one real procedure at one plant.
“Why does this need a big GPU?”	Four models live, verify on 4 frames, several trainees per GPU, all on one card.

WHO PAYS

training and safety managers

BUYER

Plant training managers and EHS managers. Training is mandatory and today costs senior technicians' hours.

PRICING

Per procedure authored, plus per trainee seat per month.

WHAT IMPROVES

Expert hours saved per trainee, mistakes caught before they happen, time to competency.

GHOST EXPERT

every plant has someone about to retire.

film them.

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